1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 HTTPS interface overview

Target secure interface `generic_https`, policy CERT_AUTH at transport; default standalone origin is `https://<provisioned-host>:8480`. The UI and API share the listener. Configure valid SANs and reverse-proxy client certificates if a proxy is used. Requests/responses are JSON except assets, echo and204login. Documented routes are the actual handlers; source registration metadata has known method/path mismatches. Tables use grouped methods or aliases where indicated.

Method	Actual handler path	Access / behavior
GET	`/health`	transport. Runtime configuration, not readiness proof
GET	`/and/ui`	transport. Static embedded index
GET	`/app.jsand/assets/app.js`	transport. Embedded JavaScript
GET	`/styles.cssand/assets/styles.css`	transport. Embedded CSS
GET	`/api/session`	console policy. authenticated/configured booleans
POST	`/api/session`	console policy. Password login; 204 and cookie
DELETE	`/api/session`	console policy. Invalidate cookie/session
GET	`/api/overview`	session. Inventory/runtime aggregation; synthetic flags noted
GET	`/api/service-registry-management`	session. Grouped devices/systems/services/runtime
GET	`/api/datamanager`	session. Flatten historian; limit default240, max1000
GET	`/api/consumers`	session. Service and simple-store consumer view
GET	`/api/intercloud`	session. Static unavailable placeholder
GET	`/api/sysmonitor`	session. Inherited observation adapter; optional system query
GET	`/api/sysmonitor/subscriptions`	session. Read subscription adapter; synthetic intake flags
PUT, DELETE	`/api/sysmonitor/subscriptions/{providerSystem}`	session + confirm. Acknowledgement only; NO provider write

Method	Actual handler path	Access / behavior
GET	`/api/plantdescriptions`	session. Plant records and registry catalogue
POST	`/api/plantdescriptions`	session + confirm. Create through provider adapter
PUT, PATCH, DELETE	`/api/plantdescriptions/{id}`	session + confirm. Provider replace/update/remove
GET	`/api/orchestration/rules`	session. Simple store plus producer/consumer catalogue
POST	`/api/orchestration/rules`	session + confirm. Validate consumer/instance/priority then create
DELETE	`/api/orchestration/rules/{id}`	session + confirm. Remove via simple-store adapter
GET	`/api/logs`	session. service query; tail default150, max500
POST	`/api/containers/{service}/{action}`	session + confirm. Static allowlist, project filter; start/stop/restart
GET	`/localcloud/management/status`	Arrowhead management. Legacy status response
GET	`/localcloud/management/systems`	Arrowhead management. Registry systems query
GET	`/localcloud/management/services`	Arrowhead management. Registry instances query
GET	`/localcloud/monitor/echo`	Arrowhead identity. Text Got it!; registry metadata differs

3 Session and mutation DTOs

`POST/api/session` body: `{"password":"..."}`. Success204with Set-Cookie `ah_console`, HttpOnly, SameSite=Strict, Path=/, Max-Age default28800; no Secure flag. `GET/api/session` responds authenticated/-configured; DELETE invalidates and expires the cookie. Missing hash503, wrong password401, lockout429with Retry-After; lockout follows five failures in five minutes and lasts one minute. Host or Origin denial403.

All `/api/` requests require allowed Host/Origin; all except session routes require a session. Mutation confirmation is exactly `X-Arrowhead-Action:confirm`; absent confirmation428. This is not cryptographic replay protection or origin-port isolation.

Rule creation body requires consumer, serviceInstanceId and positive integer priority. The adapter verifies producer availability and compatible consumer from registry data before forwarding. Successful create201; remove200with id/removed/at. Provider failures generally502with an error string; malformed rule JSON400.

Plant mutations forward JSON to the inherited provider path; create201, updates200, delete200with removed id. Subscription writes respond providerSystem/updated/removed but do not forward any change. Lifecycle responds202with service/action/accepted/at, and only start/stop/restart on a static allowlisted project container is accepted.

4 Read response DTOs

Overview includes collectedAt, cloud, summary, devices, systems, services, registryManagement, naming, infrastructure, orchestrations, containers, history, registration and errors. Grouping includes devices/runtimeOnly and registered/deployed counts. Some flags and empty collections are synthetic, not verified state.

Data view uses limit (default240,max1000), collectedAt/count/latest/entries/series; entries are flattened inherited historian data. Consumers returns localCloud/errors/rows with provider/definition/authorization/orchestration views; cloud labels are partly hard-coded TestCloud.

SysMonitor view includes collectedAt, retentionDays, storedObservations, systems, selectedSystem and observations; the optional query is system. Subscription read returns defaultSubscription/providers/subscriptions/intake, with synthetic connected/providerFound values. Plant read returns entries/systems/services/registryError.

Logs use service and tail query (default150,max500), returning service/lines/collectedAt. Runtime rows include id/name/service/displayName/image/state/status/health/created/managed. Logs may include any container returned by the Compose project filter, even when lifecycle management is disallowed.

5 Compatibility and release gates

This relocation intentionally preserves the latest located manager's business behavior. New changes are packaging (standalone nested module/role-specific launcher), embedded UI fallback, documentation and tests. This is not a security remediation or full Java-equivalence implementation.

1. `PUT/DELETE/api/sysmonitor/subscriptions/{providerSystem}` currently checks confirmation and returns updated/removed flags without forwarding a write. Do not interpret this as applied configuration. This remains a release blocker.

2. `/api/intercloud` returns a static unavailable/no-relay summary. It does not inspect Gatekeeper/GateTunnel. Consumer cloud labels include hard-coded TestCloud strings; no live foreign authorization catalogue is queried.

3. Overview hard-codes registration/naming flags and leaves several collections empty. Subscription status hard-codes intake.connected=true/providerFound=false. These are display adapters, not authoritative readiness or security evidence.

4. Historian, SysMonitor, plant and orchestration adapters target the inherited Go-cloud routes/shapes. They are not proven compatible with the newer generic Go support port or latest Java APIs. The standalone module must be qualified against the providers actually deployed.

5. Console Host/Origin checks compare host names, not exact origins/ports. The `ah_console` cookie is not port-isolated and lacks Secure. Sessions/history are in-memory and reset on restart. Missing password disables APIs. Login lockout is global, five failures in five minutes then one-minute lockout; this is not comprehensive abuse protection.

6. The manager API is dispatched before the shared application's token/management checks. Its boundary is console password/session plus the transport's mTLS if enabled, not a bearer-token-only API. Static assets and health have no console session requirement. A proxy must preserve this intended boundary.

7. Authentication has explicit CN/profile pinning. Other provider clients use CA/SAN validation but do not generally pin expected provider CN. Do not claim universal authority-substitution resistance for this inherited code. Redirects are disabled. Optional CRLs reload for new connections only; omitted CRLs mean no revocation enforcement.

8. Lifecycle uses the static systemToCompose service allowlist and Compose-project filtered container list; MANAGED_SERVICES is not consumed by this implementation. Action is start/stop/restart only. Logs, however, can read any returned project container, not just the lifecycle allowlist. The raw Docker socket still confers broad daemon privilege: application checks are not a least-privilege socket boundary. Production needs a constrained proxy and deployment review.

9. Shared runtime source includes other roles for compile-time dependencies and inherited tests. The dedicated executable refuses selecting those roles; they are not a second support-system deployment. No Docker socket or live state is needed by tests.

10. Telemetry AMQP publisher has no durable local outbox; retries/reconnects do not establish exactly-once delivery. power=line/available and process health are generated status, not measurements of physical hardware. Registration is retried every20seconds; displayed registration flags are not confirmation. No real broker test is claimed here.

11. Shared state is the inherited JSON format, not the newer support port's transactional state model. There is no new Java SQL migration, encryption-at-rest or multi-process ownership guarantee. Keep independent state and backups.

12. Historical Markdown/SysML claims of three-way interchangeability are retained only under docs/legacy. The revised model and new SysD/SysDD/SD/IDD set distinguish intended common contracts from actual Go qualification.

13. The shared serviceOperations helper generates POST/name metadata for localCloudManagement and the legacy monitor service, whereas actual handlers include GET/PUT/PATCH/DELETE and nested paths. Registration metadata is therefore not an authoritative HTTP contract. The new IDDs document actual handler methods/paths; correcting advertised metadata is separate functional work, not silently folded into relocation.

Source relocation does not modify running containers or remove their deployment source. Before switching deployment build paths, close these gaps, qualify provider DTOs/policies, review Docker access, isolate browser trust domains, and test rollback.

6 References and verification

Authoritative moved code: LocalCloudManager/internal/cloud and embedded ui sources. Exact original/relocated SHA256 hashes are in docs/source-manifest.json. docs/legacy preserves the earlier local Markdown/SysML,

not an endorsement of its parity claims. The revised SysML variability model is descriptive and has not been checked by a SysML parser. Executed checks and qualification limits are recorded in VERIFICATION.md. Source movement does not deploy or alter running clouds.

7 Revision record

2026-11-09: Documentation edition 5.3.0 derived from the retained 5.2.2 sources; metadata and local references updated. Runtime and wire-contract versions remain source-defined.